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```
#Exp No:1.a Analyze the trend of data science job postings over the last decade

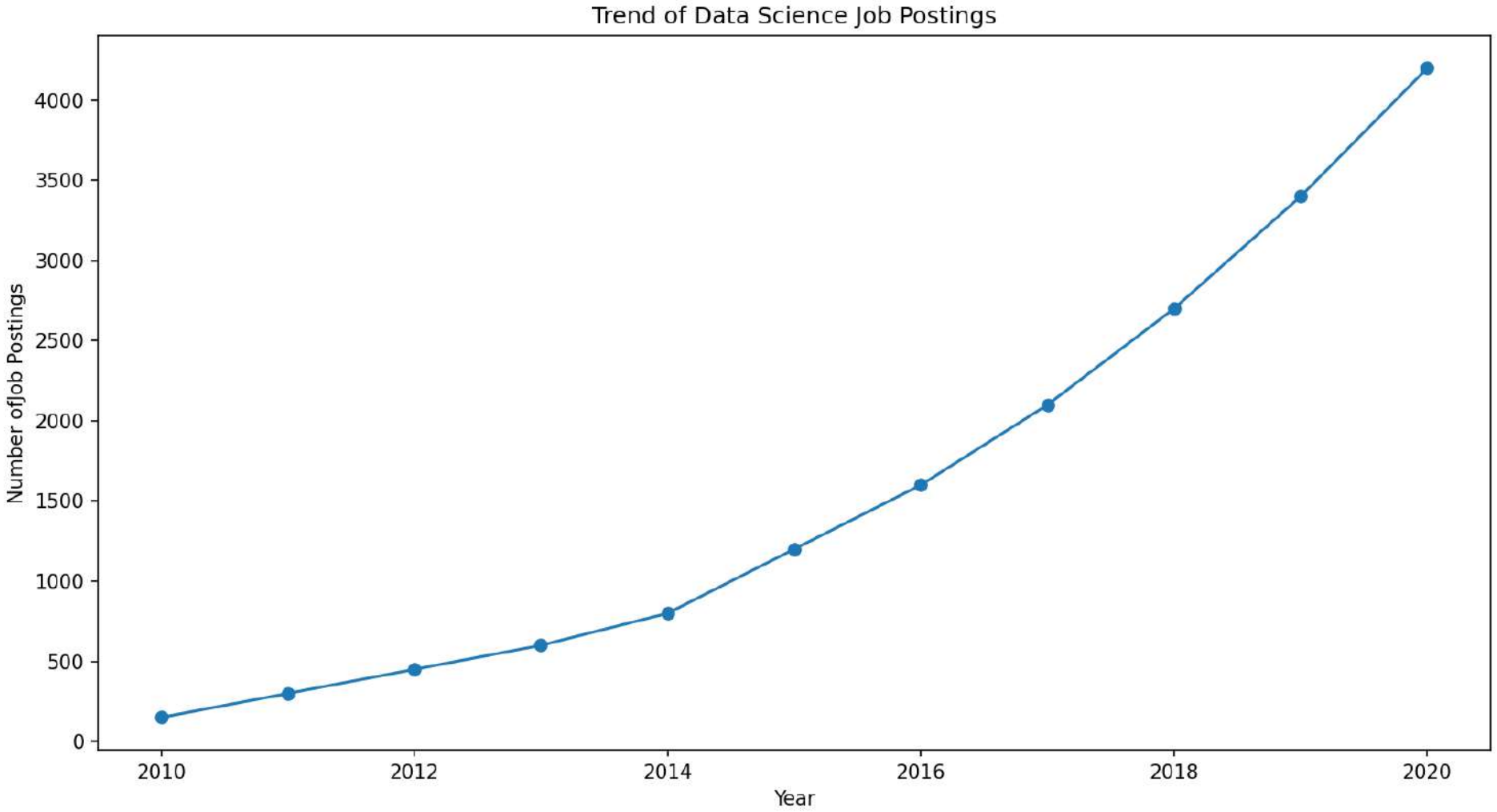
#Description: Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather
#data on the number of data science job postings each year. Use pandas for data manipulation
#and matplotlib/seaborn for visualization.

import pandas as pd
import matplotlib.pyplot as plt
data = {'Year': list(range(2010, 2021)),
'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}

df = pd.DataFrame(data)
plt.plot(df['Year'], df['Job Postings'], marker='o')
plt.title('Trend of Data Science Job Postings')
plt.xlabel('Year')
plt.ylabel('Number ofJob Postings')
plt.show()
```

Ln: 5 Col: 1

Figure 1


Trend of Data Science Job Postings
Number ofJob Postings
0
500
1000
1500
2000
2500
3000
3500
4000
2010
2012
2014
2016
2018
2020
Year

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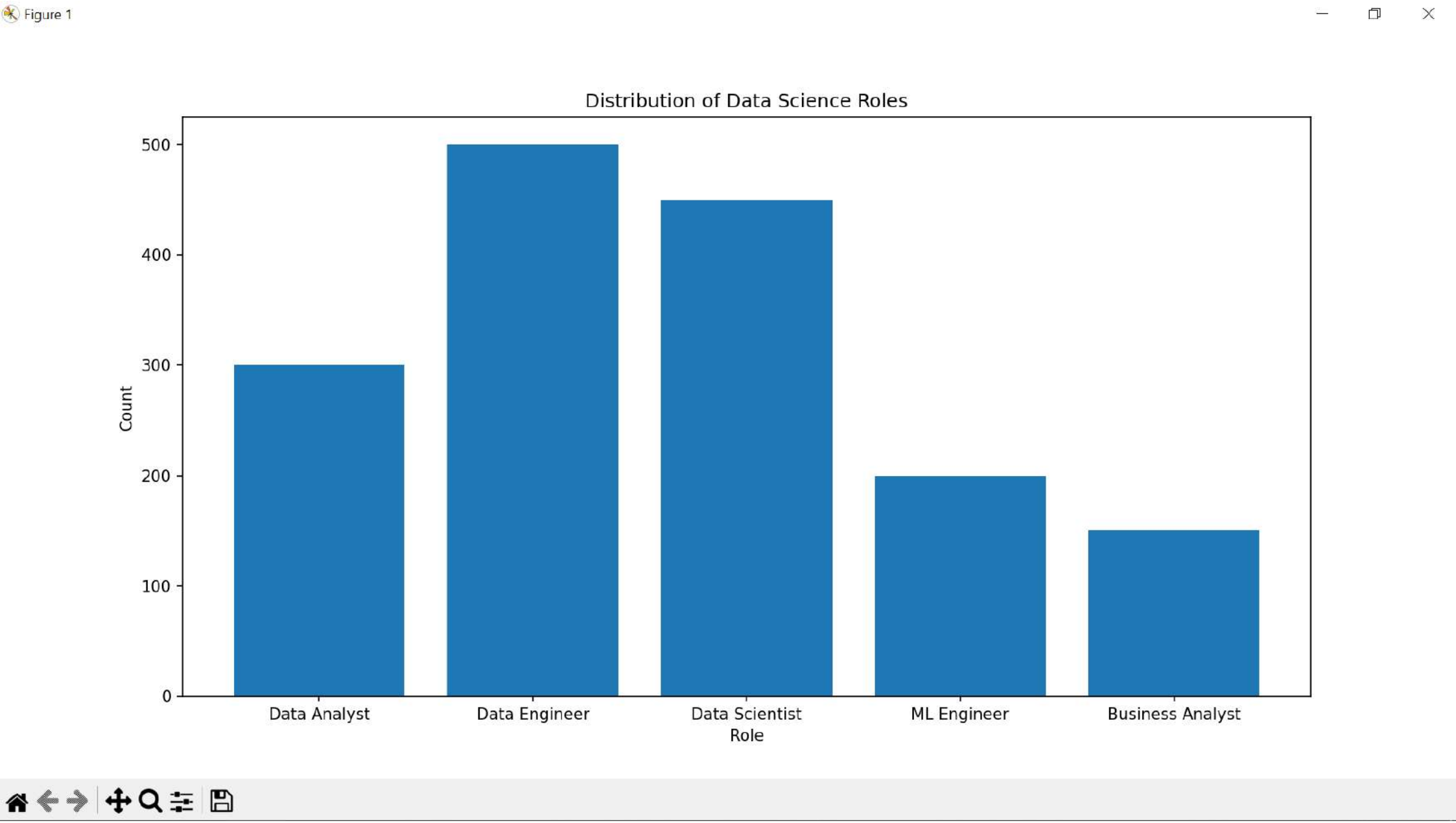
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```
#Exp No:1.b Analyze and visualize the distribution of various data science roles (Data
#Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

#Description: Use a dataset of job postings and categorize them into different roles. Visualize
#the distribution using pie charts or bar plots.

import pandas as pd
import matplotlib.pyplot as plt
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',
'Business Analyst']
counts = [300, 500, 450, 200, 150]
plt.bar(roles, counts)
plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')
plt.show()
```

Ln: 13 Col: 47


Figure 1
Distribution of Data Science Roles
Count
500
400
300
200
100
0
Data Analyst
Data Engineer
Data Scientist
ML Engineer
Business Analyst
Role

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```
#Exp No:1.c Conduct an experiment to differentiate Structured , Un-structured and Semi
#structured data based on data sets given.
#Description: Create small datasets for each type and explain their characteristics.

import pandas as pd
import matplotlib.pyplot as plt
# Structured data example
structured_data = pd.DataFrame({
'ID': [1, 2, 3],
'Name': ['Alice', 'Bob', 'Charlie'],
'Age': [25, 30, 35]
})
print("Structured Data:\n", structured_data)

# Unstructured data example
unstructured_data = "This is an example of unstructured data. It can be a piece of text, an image,or a video file."
print("\nUnstructured Data:\n", unstructured_data)

# Semi-structured data example (JSON)
semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
print("\nSemi-structured Data:\n", semi_structured_data)
```

Ln: 13 Col: 44

IDLE Shell 3.14.2

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```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/pings/Desktop/dl.py ==========
Structured Data:
    ID     Name  Age
0   1    Alice   25
1   2      Bob   30
2   3  Charlie   35

Unstructured Data:
 This is an example of unstructured data. It can be a piece of text, an image,or a video file.

Semi-structured Data:
 {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
>>>
```

Ln: 16 Col: 0

dl.py - C:/Users/pings/Desktop/dl.py (3.14.2)

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```
#Exp No:1.d Conduct an experiment to encrypt and decrypt given sensitive data.
#Description: Use the cryptography library to encrypt and decrypt a piece of data.

import pandas as pd
import matplotlib.pyplot as plt
# Generate key and encrypt data
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
b'...'
f.decrypt(token)
b'Rajalakshmi Engineering College'
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)
# Decrypt data
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

Ln: 16 Col: 0

IDLE Shell 3.14.2

File Edit Shell Debug Options Window Help

```
Python 3.14.2 (tags/v3.14.2:df79316, Dec  5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/pings/Desktop/dl.py ==========
Original Data: b'Rajalakshmi Engineering College.'
Encrypted Data: b'gAAAAABqxeO21HPKP9kGbl5EQ9GrQEhQdzEIsCOPbMcN2D2FMdO3U0Z6lOJ4CQIVwLRT5uNrMUDDmpMkQQ3OLvZ5ra_pAiMZRFzEPHdnCnvwszAFT0xUZhnAV_d9AyL4f-PEjKn
CajhF'
Decrypted Data: b'Rajalakshmi Engineering College.'
>>>
```

Ln: 8 Col: 0